

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary Summer 2024 Course: B. Tech. Branch: All Semester: II Subject Code & Name: PHY1202 Engineering Physics Max Marks: 60 Date:14/06/2024 Duration: 3 Hr.			
	Instructions to the Students: 1. Each question carries 12 marks. 2. Attempt any 5 questions from the following. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
				Marks
	Q. 1	Solve the following.		12
	A)	Explain the production of ultrasonic waves by using magnetostriction method. Calculate the length of iron rod which can be used to produce ultrasonic waves of 20 KHZ. Density of iron is $7.23 \times 10^3 \text{ kg/m}^3$, Young's modulus is $11.6 \times 10^{10} \text{ N/m}^2$		6
	B)	Define Free vibrations. Derive differential equation of free vibrations.		6
	Q.2	Solve Any Two of the following.		12
	A)	In Newton's rings, derive an expression for diameter of n^{th} bright ring and dark ring.		6
	B)	State & explain Brewster's law of Polarization. A ray of light is incident on the surface of glass plate of refractive index 1.732 at the polarizing angle. Calculate the angle of refraction of the ray.		6
	C)	Explain the construction and working of Helium -Neon laser with neat, labeled diagram.		6
	Q. 3	Solve the following.		12
	A)	Derive Schrodinger's time dependent wave equation of matter wave.		6
	B)	Explain the principle and working of Bainbridge mass Spectrograph.		6
	Q.4	Solve the following questions.		12

A)	Calculate atomic packing factor for Simple cubic, Body centered cubic and Face centered Cubic structure.		6
B)	Distinguish between Characteristics and continuous X-ray spectra.		6
Q. 5	Solve the following questions.		12
A)	Explain the classification of magnetic materials.		6
B)	Distinguish between Type I and Type II superconductors.		6
Q. 6	Solve the following questions.		12
A)	Derive an equation for conductivity of semiconductors.		6
B)	Derive an expression for electromagnetic wave in free space. Hence calculate the velocity of light in free space.		6
	*****End*****		